

Lab3: Logistic Regression

2540002-Nguyen Nhat Anh

May 2026

1 Introduction

This report represents the code and the flow for the lab work and the logic implementation to the code of this lab work.

2 Recap Mathematics Requirements

$$f(w_0, w_1, w_2) = -y_i(w_1x_i^{(1)} + w_2x_i^{(2)} + w_0) + \log(1 + e^{w_1x_i^{(1)} + w_2x_i^{(2)} + w_0}) \quad (18)$$

$$\begin{aligned} \frac{df}{dw_0} &= -y_i + \frac{e^{w_1x_i^{(1)} + w_2x_i^{(2)} + w_0}}{1 + e^{w_1x_i^{(1)} + w_2x_i^{(2)} + w_0}} \\ &= 1 - y_i - \frac{1}{1 + e^{w_1x_i^{(1)} + w_2x_i^{(2)} + w_0}} \\ &= 1 - y_i - \sigma(-(w_1x_i^{(1)} + w_2x_i^{(2)} + w_0)) \end{aligned}$$

$$\begin{aligned} \frac{df}{dw_1} &= -y_ix_i^{(1)} + \frac{x_i^{(1)}e^{w_1x_i^{(1)} + w_2x_i^{(2)} + w_0}}{1 + e^{w_1x_i^{(1)} + w_2x_i^{(2)} + w_0}} \\ &= -y_ix_i^{(1)} + x_i^{(1)}\left(1 - \frac{1}{1 + e^{w_1x_i^{(1)} + w_2x_i^{(2)} + w_0}}\right) \\ &= -y_ix_i^{(1)} + x_i^{(1)}(1 - \sigma(-(w_1x_i^{(1)} + w_2x_i^{(2)} + w_0))) \end{aligned} \quad (21)$$

$$\begin{aligned}
\frac{df}{dw_2} &= -y_i x_i^{(2)} + \frac{x_i^{(2)} e^{w_1 x_i^{(1)} + w_2 x_i^{(2)} + w_0}}{1 + e^{w_1 x_i^{(1)} + w_2 x_i^{(2)} + w_0}} \\
&= -y_i x_i^{(2)} + x_i^{(2)} \left(1 - \frac{1}{1 + e^{w_1 x_i^{(1)} + w_2 x_i^{(2)} + w_0}}\right) \\
&= -y_i x_i^{(2)} + x_i^{(2)} (1 - \sigma(-(w_1 x_i^{(1)} + w_2 x_i^{(2)} + w_0)))
\end{aligned} \tag{23}$$

3 Implementation

The code implemented by the following idea:

- Step 1: Random initialization $w_0 = 0, w_1 = 1, w_2 = 2$
- Step 2: descent...
 - $w_0 = w_0 - r * \frac{dL}{dw_0}$
 - $w_1 = w_1 - r * \frac{dL}{dw_1}$
 - $w_2 = w_2 - r * \frac{dL}{dw_2}$
- Step 3: compute $f(w_0, w_1, w_2)$. Still big?
 - Move to point (w_0, w_1, w_2)
 - Repeat Step 2

To test the code, we test on threshold = 0.1 with learning rate = 0.1, using "loan2.csv" data.

4 Result

The output is shown as:

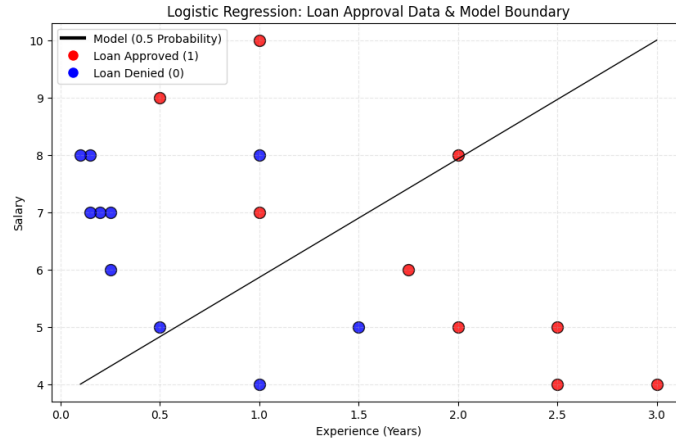


Figure 1: Linear Regression Result

5 Discussion

The learning rate = 0.1 and threshold = 0.1 shows the perfect region of the data. Also, by limiting the range of salary and the experience data from the file, the region can be clearly shown the region from the output. In each region, there are 4 errors for the "Denied" region and 2 errors for the "Approved" region.